Lecture 18
Sampling Distributions and the Standard Error

Review

1 What is a Poisson random variable?

The number of events in a fixed interval, when events occur independently at an average rate λ.

2 What is a probability density?

How concentrated probability is around a point; probability is the area under the curve.

3 For a continuous random variable, what is P(X = c)?

0: a single value has no width, so it has no area under the curve.

4 What is a cumulative probability?

$P(X \leq x)$: the area to the left of x, what the z-table and pnorm give.

Two polls, two answers

Two polling firms each survey a random sample of 1,000 voters in the same state, in the same week.

Poll A: $\hat{p} = 0.52$ Poll B: $\hat{p} = 0.49$

(a) Is one of the polls wrong?

(b) If 1,000 firms each polled 1,000 voters, what would their 1,000 values of $\hat{p}$ look like?

(a) Not necessarily: different random samples give different values of $\hat{p}$

(b) A pile of values centered near p: the sampling distribution of $\hat{p}$

Three distributions

population distribution

how a single observation X varies; described by parameters such as μ, σ and p

data distribution

the n values in our one sample; described by statistics such as $\bar{x}$, s and $\hat{p}$

sampling distribution

how a statistic such as $\bar{x}$ varies over all possible samples of the same size n

Building a sampling distribution

A spinner lands on 1, 2 or 3, each with probability 1/3. Spin it twice and record the mean, $\bar{x}$.

$\mu = 2$; n = 2 spins; $3 \times 3 = 9$ equally likely samples (order matters)

(a) List the 9 samples and the mean of each.

(b) What is $P(\bar{x})$ for each possible value of $\bar{x}$?

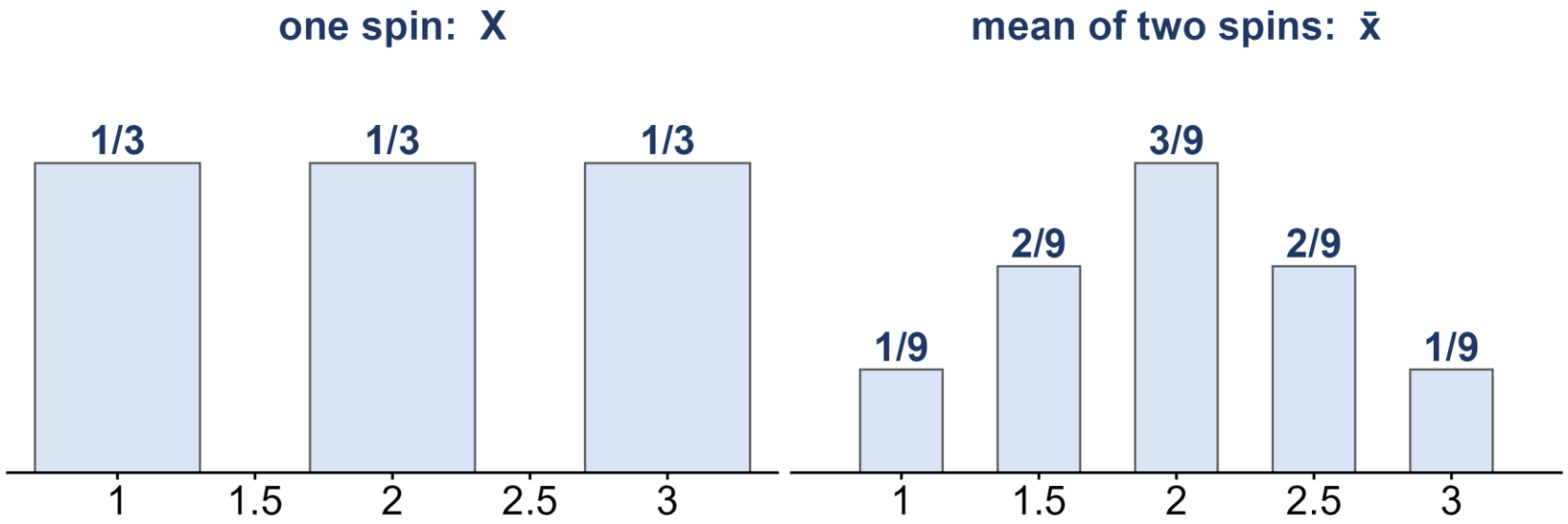
(a) (1,1) → 1; (1,2), (2,1) → 1.5; (1,3), (2,2), (3,1) → 2; (2,3), (3,2) → 2.5; (3,3) → 3

(b) $\bar{x}$ = 1, 1.5, 2, 2.5, 3 with $P(\bar{x})$ = 1/9, 2/9, 3/9, 2/9, 1/9

One spin vs. the mean of two spins

The population (left) and the sampling distribution of $\bar{x}$ for n = 2 (right).

What stayed the same? What changed?



Same center: both have mean 2. Smaller spread: $\sigma = 0.82$, $\sigma_{\bar{x}} = 0.58 = 0.82 / \sqrt{2}$
New shape: peaked in the middle

Center and spread of $\bar{x}$

mean of $\bar{x}$

$\mu_{\bar{x}} = \mu$: $\bar{x}$ is centered at the parameter it estimates

standard error of $\bar{x}$

$\sigma_{\bar{x}} = \sigma / \sqrt{n}$: the spread of $\bar{x}$, which shrinks as n grows

the sample proportion $\hat{p}$

mean p and standard error $\sigma_{\hat{p}} = \sqrt{(p(1 - p) / n)}$

Data: Spotify songs

Spotify and YouTube statistics for the top 10 songs of each of 2,079 popular artists, collected in February 2023.

X = the length of a song, in minutes

TRACK	ARTIST	MINUTES
Feel Good Inc.	Gorillaz	3.71
Rhinestone Eyes	Gorillaz	3.34
New Gold (feat. Tame Impala and Bootie Brown)	Gorillaz	3.59
On Melancholy Hill	Gorillaz	3.90
Clint Eastwood	Gorillaz	5.68
…	…	…

18,860 different songs (a song credited to several artists is counted once)

Example: standard error and n

Treat the 18,860 songs as the population. Draw a random sample of n songs and average their lengths to get $\bar{x}$. Every new sample gives a different $\bar{x}$; the standard error $\sigma_{\bar{x}}$ measures how much $\bar{x}$ varies from sample to sample.

μ = 3.71 min; σ = 1.51 min

(a) What is $\sigma_{\bar{x}}$ for n = 20?
(b) What is $\sigma_{\bar{x}}$ for n = 80?
(c) How much larger must a sample be to cut the standard error in half?

(a) 1.51 / √20 ≈ 0.34 min (b) 1.51 / √80 ≈ 0.17 min
(c) 4 times larger, because √4 = 2